

POWERPOINT PRESENTATION CHEAT SHEET

Simulating Random Graph Models in Python

10-Minute Interim Presentation

OVERALL TIMING & STRUCTURE

Time	Slide	Section	Duration
0:00	1	Title slide	15 sec
0:15	2	Motivation	1:15
1:30	3	Erdős–Rényi definition	1:30
3:00	4	ER structural interpretation	1:00
4:00	5	Watts–Strogatz construction	1:15
5:15	6	WS clustering & path length	1:15
6:30	7	Barabási–Albert construction	1:00
7:30	8	BA power law & scale-free	1:15
8:45	9	Comparison table	0:45
9:30	10	Implications & conclusion	0:30
10:00		Questions	

Golden Rule: If you reach slide 7 (BA) by 7:30, you're perfectly on track.

SLIDE 1: TITLE SLIDE (15 seconds)

What's on the slide:

- Title: Simulating Random Graph Models in Python
- Your name
- Department/University info
- Date

What to say:

"Good morning/afternoon. Today I'll present the mathematical foundations of three principal random graph models and their structural properties. This is interim work from my dissertation project."

Notes:

- Don't linger — this is just housekeeping

- Make eye contact, then advance immediately
 - Establish confident tone from the start
-

SLIDE 2: MOTIVATION (1:15 — arrive 0:15, leave 1:30)

What's on the slide:

- **Why Study Random Graph Models?**
- Networks are everywhere (social, internet, neural, citation)
- Key questions: How do networks form? Why short paths? Why hubs?
- Goal: Understand mathematical foundations through theory and simulation

What to say (memorize these bullets):

1. Opening statement:

"Networks are fundamental structures across disciplines — from social networks to the internet to neural connectivity."

2. The puzzle:

"Real-world networks exhibit properties that deterministic models cannot capture. Why do most networks have short average path lengths — the small-world property? Why do a few nodes attract most connections?"

3. Your approach:

"My project addresses these questions by studying three foundational random graph models: their mathematical definitions, structural properties, and the phenomena they explain."

4. Transition:

"Let's begin with the oldest and most studied model — Erdős–Rényi."

Delivery tips:

- Speak slowly on the opening sentence — set the pace
 - Gesture to the bullet points but don't read them verbatim
 - The "transition" sentence is your bridge — memorize it exactly
-

SLIDE 3: ERDŐS–RÉNYI DEFINITION (1:30 — arrive 1:30, leave 3:00)

What's on the slide:

- **The Erdős–Rényi Model $G(n,p)$**
- Definition: n vertices, each edge independently with probability p
- Expected edges: $E[|E|] = p \cdot C(n,2)$
- Degree distribution: $d(v) \sim \text{Binomial}(n-1, p)$
- Expected degree: $E[d(v)] = p(n-1)$
- Connectivity threshold: $p_c = (\ln n) / n$

What to say (section by section):

Definition (30 sec):

"The Erdős–Rényi model $G(n,p)$ is defined on n vertices where each possible edge is included independently with probability p . The key word is 'independently' — there's no memory, no preference, no structure."

Expected edges (15 sec):

"There are n -choose-2 possible edges, each appears with probability p , so the expected number of edges is p times n -choose-2."

Degree distribution (30 sec):

"The degree of any vertex is binomially distributed. Why? Because each of the other $n-1$ vertices contributes independently to the degree. The expected degree is simply p times $n-1$."

Connectivity threshold (30 sec):

"One of the most beautiful results: there's a sharp threshold at p equals $\ln n / n$. Below this, isolated vertices persist almost surely. Above it, the graph becomes connected with high probability. This is a phase transition — a tiny change in p causes a massive structural shift."

Transition (5 sec):

"But what does this model actually produce structurally?"

Delivery tips:

- Point to each formula as you explain it
- Emphasize "independently" twice — it's the model's defining characteristic
- When saying " $p_c = \ln(n)/n$ ", trace the formula with your finger
- Pause after "phase transition" — let it land

SLIDE 4: ER STRUCTURAL INTERPRETATION (1:00 — arrive 3:00, leave 4:00)

What's on the slide:

- **Erdős–Rényi: Structural Interpretation**
- Phase transitions: below $p_c \rightarrow$ disconnected; at $1/n \rightarrow$ giant component; above $p_c \rightarrow$ connected
- Limitations: Binomial degree (no heavy tails), Low clustering ($C \approx p$)
- Alertbox: "Elegant mathematics, but fails to capture clustering and heavy-tailed distributions"

What to say:

Phase transition recap (20 sec):

"Below the threshold: many small disconnected components. At p equals one over n , a giant component of size proportional to n suddenly emerges. Above the threshold, the graph is almost

surely fully connected."

Limitations (30 sec):

"As a model of real networks, ER has critical flaws. The degree distribution is binomial — tightly concentrated around the mean with no heavy tails. No hubs. And clustering is approximately p , which for sparse graphs is tiny. Real social networks have high clustering — ER doesn't capture this at all."

Key takeaway (10 sec):

"Erdős–Rényi gives us sharp thresholds and rigorous probabilistic analysis, but the resulting graphs are homogeneous and structureless — nothing like reality."

Transition:

"This leads us to ask: can we build a model with local structure while maintaining short paths? That's Watts–Strogatz."

Delivery tips:

- Build suspense on "critical flaws"
- When listing limitations, count them on your fingers (1: no tails, 2: no clustering)
- The alertbox is the punchline — read it clearly

SLIDE 5: WATTS–STROGATZ CONSTRUCTION (1:15 — arrive 4:00, leave 5:15)

What's on the slide:

- **The Watts–Strogatz Small-World Model**
- Construction:
 1. Begin with regular ring lattice (n vertices, each connected to k nearest neighbors)
 2. Rewire each edge with probability β to random vertex
- Effect of β :
 - $\beta = 0$: pure lattice (high clustering, large path length)
 - $\beta = 1$: effectively random (low clustering, short path length)
 - $0 < \beta < 1$: small-world regime (high clustering + short paths)

What to say:

Construction step 1 (20 sec):

"Watts and Strogatz start with a structured foundation: a regular ring lattice where n vertices form a circle, and each vertex connects to its k nearest neighbors. This is completely ordered — maximum local structure."

Construction step 2 (20 sec):

"Now introduce randomness: rewire each edge independently with probability β to a uniformly chosen random vertex. We're not adding edges, we're redirecting them."

The β parameter (30 sec):

"Beta is the control knob. At beta equals zero, we have the pure lattice: high clustering, but you have to traverse half the ring to reach distant nodes — inefficient. At beta equals one, we've effectively randomized everything — low clustering, but short paths like Erdős–Rényi. The magic happens in between."

Small-world regime (15 sec):

"For small beta — say 0.01 or 0.05 — most edges stay in place, preserving local triangles and clustering. But just a handful of random shortcuts act as bridges, dramatically reducing average path length."

Transition:

"Let's see exactly why this works."

Delivery tips:

- Use hand gestures: circular motion for "ring," cutting motion for "shortcuts"
- Emphasize "control knob" — β is the single tuning parameter
- Slow down on "The magic happens in between" — this is the core insight

SLIDE 6: WS CLUSTERING & PATH LENGTH (1:15 — arrive 5:15, leave 6:30)**What's on the slide:**

- **Watts–Strogatz: Clustering and Path Length**
- Clustering coefficient = probability that two neighbors of a vertex are themselves connected
- Why clustering stays high: most edges remain in place; lattice structure preserved
- Why path length drops: even few shortcuts bridge distant regions
- Reduces diameter from $O(n/k)$ to $O(\log n)$
- Alertbox: "High clustering + short paths coexist — the small-world phenomenon"

What to say:**Clustering coefficient (20 sec):**

"The clustering coefficient measures local cohesion — informally, if your two friends are likely to be friends with each other. In a lattice, this is high because neighbors form tight triangles. With small beta, we don't destroy many triangles, so clustering remains high."

Path length reduction (30 sec):

"But here's the key: even a tiny fraction of rewired edges — say one percent — creates shortcuts that span large distances across the ring. These shortcuts collapse the diameter. On the pure lattice, average path length is order n over k — proportional to network size. With shortcuts, it drops to order $\log n$ — like a random graph."

The phenomenon (15 sec):

"This is the small-world phenomenon: you can have both high local clustering — like a regular lattice — and short global paths — like a random graph. These aren't opposites. They coexist precisely in the range where beta is small but nonzero."

Real-world relevance (10 sec):

"This regime captures the structure of social networks, neural networks, and many other real-world systems where local communities exist but the world still feels small."

Transition:

"But Watts–Strogatz still doesn't explain hubs. For that, we need growth and feedback — enter Barabási–Albert."

Delivery tips:

- Define clustering coefficient simply — don't get bogged down in formulas
- When saying "shortcuts collapse the diameter," gesture broadly to illustrate spanning distance
- Emphasize "coexist" — this is the conceptual breakthrough
- The transition should build anticipation for BA

SLIDE 7: BARABÁSI–ALBERT CONSTRUCTION (1:00 — arrive 6:30, leave 7:30)

What's on the slide:

- **The Barabási–Albert Preferential Attachment Model**
- Construction:
 1. Begin with small initial graph (m_0 vertices)
 2. At each time step: add one new vertex with m edges
 3. Preferential attachment: $P(i) = d_i / \sum_j d_j$
- Intuition: "The rich get richer"

What to say:

Growth mechanism (20 sec):

"Unlike the previous models, Barabási–Albert is dynamic — the network grows over time. We start with a small initial graph, then at each step we add one new vertex along with m new edges."

Preferential attachment (30 sec):

"Here's the critical rule: the new edges don't connect uniformly at random. Instead, the probability of connecting to an existing vertex i is proportional to i 's current degree. High-degree nodes attract more connections. This is preferential attachment."

The formula (15 sec):

"Mathematically: the probability of connecting to vertex i equals d_i divided by the sum of all degrees. This ratio ensures probabilities sum to one and encodes the feedback: popularity breeds popularity."

Intuition (10 sec):

"We call this 'the rich get richer' or the Matthew effect. Nodes that get an early advantage continue to accumulate connections faster than others."

Transition:

"This simple mechanism has profound consequences for the degree distribution."

Delivery tips:

- Emphasize "dynamic" — this is the first growth model
- Point directly at the formula $P(i) = d_i / \sum d_j$ when explaining it
- Use the phrase "the rich get richer" — it's memorable and intuitive
- Gesture upward when saying "accumulate connections faster"

SLIDE 8: BA POWER LAW & SCALE-FREE (1:15 — arrive 7:30, leave 8:45)**What's on the slide:**

- **Barabási–Albert: Scale-Free Networks**
- Power-law degree distribution: $P(k) \sim k^{-3}$
- Derived rigorously via mean-field / master equation
- What power law implies structurally:
 - Heavy tail: few vertices (hubs) accumulate disproportionate edges
 - Scale-free: no characteristic degree scale
 - Robustness to random failure, vulnerable to targeted attack
- Alertbox: "Preferential attachment generates heavy-tailed distributions observed in web, citations, internet"

What to say:**Power law (25 sec):**

"Preferential attachment produces a power-law degree distribution: P of k goes as k to the minus three. This isn't a fitted parameter — it's a theoretical prediction derived from the growth dynamics using mean-field analysis or master equations."

Heavy tail (20 sec):

"Power laws have heavy tails. Most vertices have low degree — one or two connections. But a small number of hubs have enormous degree — hundreds or thousands of connections. This is fundamentally different from the binomial distribution in Erdős–Rényi, which has exponential decay."

Scale-free (20 sec):

"We call these networks 'scale-free' because there's no characteristic degree scale. If you zoom in or out, the distribution looks the same — it's self-similar. The power law has no finite variance in the limit, meaning hubs can be arbitrarily large."

Robustness paradox (15 sec):

"Hubs create a structural paradox: the network is robust to random node failures because most nodes are low-degree and redundant. But it's fragile under targeted attacks — remove the hubs and the network disintegrates."

Real-world examples (10 sec):

"This structure appears everywhere: the web graph, citation networks, protein interactions, the internet backbone. Barabási–Albert explains why a few websites, papers, or proteins dominate their respective networks."

Transition:

"So we have three models. Let's compare them directly."

Delivery tips:

- Write " $\sim k^{-3}$ " in the air with your finger when saying it
- Contrast "most have low degree" (gesture low) with "hubs have enormous degree" (gesture high)
- When saying "scale-free," emphasize that this is a technical term, not just metaphor
- The robustness paradox is subtle — slow down and enunciate clearly

SLIDE 9: COMPARISON TABLE (0:45 — arrive 8:45, leave 9:30)**What's on the slide:**

- **Comparative Structural Properties**
- Table with columns: ER, WS, BA
- Rows: Mechanism, Degree distribution, Clustering, Hubs, Typical uses

What to say:**Setup (10 sec):**

"Here's a direct comparison across the three models. Each captures a different structural feature."

Row-by-row (30 sec):

"Mechanism: ER is pure randomness, WS rewires a lattice, BA grows with preferential attachment. Degree distribution: ER is binomial — concentrated. WS is approximately Poisson for small beta — still concentrated. BA is power law — heavy-tailed with hubs. Clustering: ER is low, roughly p . WS is high for small beta. BA is low to moderate. Hubs: only BA produces them. Typical applications: ER for percolation and thresholds, WS for social and neural networks, BA for the web and scale-free systems."

Synthesis (10 sec):

"No single model explains everything. Each isolates one mechanism. Real networks often require hybrid models or more sophisticated approaches."

Transition:

"Let me conclude with the broader implications."

Delivery tips:

- Don't linger — this slide is a summary, not new content
- Point to each cell as you mention it (laser pointer if available)
- Speak slightly faster here — you're running low on time
- The synthesis sentence is important: memorize it verbatim

SLIDE 10: IMPLICATIONS & CONCLUSION (0:30 — arrive 9:30, leave 10:00)

What's on the slide:

- **Theoretical Implications and Conclusions**
- ER: exact thresholds, but too homogeneous
- WS: local structure + global efficiency are compatible
- BA: growth + feedback → heavy tails
- Limitations: WS no power law, BA low clustering, all assume undirected/unweighted
- Summary: Each model captures distinct facets; together they motivate richer theory

What to say:

Theoretical implications (20 sec):

"What have we learned? Erdős–Rényi gives us exact threshold results but produces unrealistic homogeneous graphs. Watts–Strogatz shows that local clustering and global efficiency aren't mutually exclusive — a key insight. Barabási–Albert demonstrates that growth and feedback alone are sufficient to generate empirically observed heavy-tailed distributions."

Limitations (10 sec):

"Each model has limitations. Watts–Strogatz doesn't produce power laws. Barabási–Albert has low clustering — hybrid models like Holme–Kim address this. All three assume undirected, unweighted graphs — real networks are often directed and weighted."

Closing statement (5 sec):

"Together, these models form the foundation of network science and motivate ongoing research into more refined generative mechanisms. Thank you — I'm happy to take questions."

Delivery tips:

- This is your landing — stick it cleanly
- Don't rush the closing statement even if time is tight
- Make eye contact on "Thank you"
- Pause for 2 seconds before saying "questions" to let it settle

ANTICIPATED QUESTIONS & ANSWERS

Q1: "What happens exactly at the threshold p_c in Erdős–Rényi?"

Answer:

"At $p_c = \ln(n)/n$, the giant component emerges — a connected component containing a constant fraction of all vertices. Below p_c , all components are logarithmic in size. At p_c , we're at the critical point of a phase transition. The structure changes discontinuously as we cross the threshold."

Q2: "Why is the BA exponent exactly -3 ?"

Answer:

"The exponent -3 comes from solving the master equation for how degree evolves over time. When you assume linear preferential attachment — probability proportional to degree — and solve the rate equation, you get a power law with exponent -3 . It's not a fit; it's a theoretical prediction. If you change the attachment rule to nonlinear — say proportional to degree squared — you get a different exponent."

Q3: "Can Watts–Strogatz produce a power-law degree distribution?"

Answer:

"No. The degree distribution in WS remains approximately Poisson for small beta and becomes more random-graph-like as beta increases. But it never develops heavy tails. WS preserves or slightly perturbs the original lattice degree, so you don't get hubs. If you want power laws, you need a growth mechanism like BA or a fitness model."

Q4: "How does BA handle clustering?"

Answer:

"It doesn't, really. Clustering in pure BA is low — around order $1/n$, which vanishes as the network grows. This is a known limitation. Hybrid models address it. For example, the Holme–Kim model adds a triangle-closing step after each preferential attachment: with some probability, you close a triangle instead of attaching to a random high-degree node. This increases clustering while preserving the power-law degree distribution."

Q5: "What is the clustering coefficient formally?"

Answer:

"For a vertex v , it's the ratio of actual triangles containing v to the maximum possible number of triangles. If v has degree k , there are k -choose-2 possible edges between v 's neighbors. The clustering coefficient is the number of those edges that actually exist, divided by k -choose-2. Averaging over all vertices gives the global clustering coefficient."

Q6: "Are these models realistic for modern networks?"

Answer:

"Each captures one important feature, but real networks are more complex. For example, social networks have community structure that none of these models explicitly generate. The internet has geographic and administrative constraints. Citation networks have temporal dynamics beyond simple growth. So we use these as building blocks, but modern research often combines multiple mechanisms — growth, clustering, community structure, fitness distributions — to better match empirical data."

Q7: "What about directed graphs?"

Answer:

"All three models as I've presented them are undirected. You can extend BA to directed graphs by distinguishing in-degree and out-degree, and making preferential attachment depend on in-degree — this models citation networks well. ER extends trivially: just include each directed edge with probability p . WS is less commonly studied in the directed case, though you could define a directed ring lattice and rewire directed edges."

Q8: "How do you choose p or β or m in practice?"

Answer:

"It depends on what you're modeling. In ER, p is often chosen to match a target edge density or to explore the threshold regime. In WS, β is typically set small — 0.01 to 0.1 — to stay in the small-world regime. In BA, m controls the minimum degree: every new vertex has degree m , so larger m gives denser networks. You'd match these parameters to empirical network properties like average degree or clustering."

Q9: "Can you combine these models?"

Answer:

"Yes, absolutely. There are many hybrid models. For instance, you could grow a network using BA preferential attachment but on a WS-like substrate to inject geographic or spatial constraints. Or use a fitness model where each vertex has an intrinsic attractiveness in addition to degree-based preference. Combining mechanisms is an active area of research — the goal is to capture multiple real-world properties simultaneously."

Q10: "What about weighted edges?"

Answer:

"None of these models assign weights to edges as constructed. But you could extend them. For example, in a weighted BA model, you might make attachment probability proportional to the sum of edge weights instead of degree. In WS, after rewiring you could assign weights based on geographic distance if vertices have spatial positions. Weighted networks are more complex but often more realistic."

COMMON MISTAKES TO AVOID

Content mistakes:

- **Don't** confuse the two ER models ($G(n,m)$ vs $G(n,p)$) — you're using $G(n,p)$
- **Don't** say WS produces power laws — it doesn't
- **Don't** claim BA always has high clustering — it has low clustering
- **Don't** mix up "clustering coefficient" with "clustering" (informal usage)

Presentation mistakes:

- **Don't** read slides verbatim — slides are prompts, not scripts
- **Don't** turn your back to the audience to read the screen
- **Don't** skip the comparison table — it's the synthesis slide
- **Don't** go over 10 minutes — questions are part of the assessment
- **Don't** apologize ("sorry this is rushed" / "I didn't have time to...") — own your work

Body language mistakes:

- **Don't** stand still like a statue — move naturally but don't pace
- **Don't** put hands in pockets — use them to gesture
- **Don't** avoid eye contact — scan the room, make brief eye contact with individuals
- **Don't** speak to the screen — face the audience

PRESENTATION CHECKLIST

The night before:

- ☐ Read through the cheat sheet twice
- ☐ Memorize the key insight sentence for each model
- ☐ Memorize all four transition sentences
- ☐ Rehearse the comparison table bullet points (30-second version)
- ☐ Practice answering all 10 questions out loud
- ☐ Time yourself once through the full presentation (should be 9:30–10:00)

One hour before:

- ☐ Review the timing breakdown
- ☐ Mentally walk through slides 3, 4, 7, 8 (the dense ones)
- ☐ Confirm you can define clustering coefficient in one sentence
- ☐ Confirm you can state preferential attachment formula correctly

Waiting outside the room:

- ☐ Take three deep breaths
- ☐ Visualize nailing the opening sentence
- ☐ Remind yourself: "I know this material better than anyone in the room"

While presenting:

- ☐ Check watch at slide 3 (should be ~3:00), slide 7 (should be ~7:30)
- ☐ If running over: cut one bullet point from slide 4 or 6, never cut comparison
- ☐ Speak to the back row — project your voice
- ☐ Pause after each major formula or result — let it sink in

During questions:

- ☐ Listen to the full question before answering
- ☐ If unsure: "That's a good question. Let me think..." (buy 5 seconds)
- ☐ If you don't know: "I haven't explored that aspect yet, but it's an interesting direction"
- ☐ Bridge back to what you do know: "What I can tell you is..."

FINAL TIPS FOR SUCCESS

1. **Energy management:** Start strong, maintain energy through ER and WS, then push through BA even if tired.
2. **Vocal variety:** Vary your pace. Speed up slightly on slide 9 (comparison), slow down on key insights.
3. **Emphasis words:** "independently" (ER), "coexist" (WS), "rich get richer" (BA), "heavy-tailed" (BA)
4. **If you lose your place:** Glance at the slide title, take a breath, paraphrase what's on screen. Never freeze.
5. **If time runs out:** Skip to slide 10, say "To conclude..." and deliver the closing statement. Never just stop mid-sentence.
6. **Confidence projectors:**
 - Stand tall, shoulders back
 - Speak at 80% of your maximum volume (feels loud, sounds confident)
 - Make deliberate gestures (not fidgeting)
 - Smile once at the start, once at the end
7. **The secret:** They want you to succeed. They're not trying to catch you out. They're interested in the mathematics. Teach them something and you'll do well.

EMERGENCY PROTOCOLS

If the projector fails:

- Remain calm: "I'll present without slides while we troubleshoot."
- Describe each model verbally in the same order
- Draw $G(n,p)$, ring lattice, and hub diagram on the board if available
- Time limit still applies — don't let tech issues throw you

If you blank on a formula:

- Describe it in words: "The expected degree is the connection probability times the number of possible neighbors"
- Move on quickly — one formula isn't worth 30 seconds of silence

If someone asks a question you truly can't answer:

- "That's outside the scope of my current work, but I'd be happy to explore it."
- Or: "I haven't come across that in my reading yet — do you have a reference I could look at?"
- Never guess or make up an answer

If you finish at 9:00 (one minute early):

- Don't panic, don't apologize
- Say: "I'm happy to elaborate on any of these models if there's interest, or take questions now."
- One minute early is better than one minute late

If you finish at 11:00 (one minute over):

- This is not ideal but not disastrous
- Questions may be slightly shortened
- Don't mention it or apologize
- In future rehearsals, identify 30-60 seconds to cut (usually slide 4 or 6)

WHAT SUCCESS LOOKS LIKE

You will know you did well if:

- You covered all three models with their key mathematical properties
- You clearly stated the degree distributions (Binomial, $\sim$ Poisson, Power law)
- You explained clustering in WS and hubs in BA
- You completed the comparison table
- You stayed within 10 minutes $\pm$ 30 seconds
- You answered at least one question confidently
- You maintained composure even when nervous

Don't worry if:

- You stumbled over one or two words
- You had to glance at your notes once or twice
- One slide took 20 seconds longer than planned (as long as you compensated elsewhere)
- You didn't answer one question perfectly (no one expects perfection)

Red flags (avoid these):

- Going significantly over time ($> 10:30$)
- Skipping an entire model
- Misdefining a core concept (e.g., saying WS has power law)
- Freezing for > 10 seconds
- Arguing with the audience during questions

POST-PRESENTATION REFLECTION

After you finish, jot down:

1. What went well?
2. What would you do differently?
3. What question caught you off guard?
4. What slide took longer than expected?

Use this to improve for the final presentation.

YOU'VE GOT THIS.

You know the mathematics. You understand the models. You've prepared thoroughly.

Walk in, teach them something, walk out. Simple as that.

Good luck.